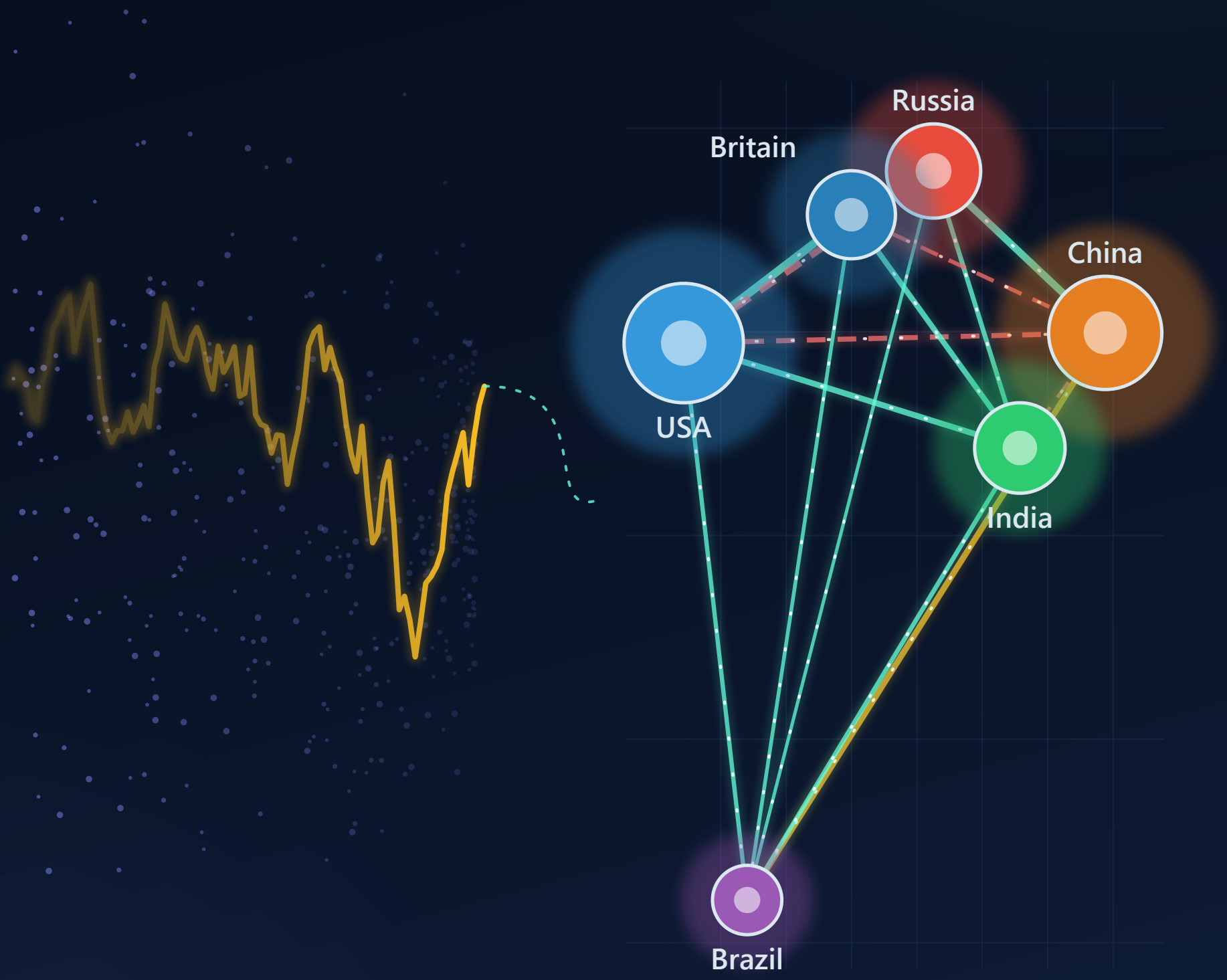
Do whales leave signatures in technical indicators?

I built a network to find out. Then I changed the question.



price: signal dissolving into noise (illustration)

A network that learns its own indicator settings

Gradients from the losses reach the periods themselves.

OHLCV window — `60 bars of 1-minute bars`

↓

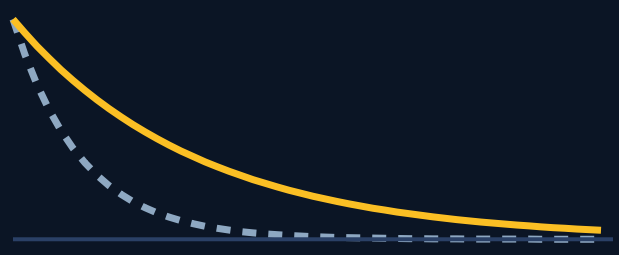
Learnable indicators

kernel width is learned

14 families x 3 instances, 54 learnable periods

```
alpha = sigmoid(logit)
period = 2/alpha - 1
```

an exponential smoother is a 1-D convolution



gradients flow back to the periods

↓

Bi-GRU → **attention, 8 heads**

↓

horizon 10 min
- price change
- P(up)
- variance

horizon 15 min
- price change
- P(up)
- variance

horizon 20 min
- price change
- P(up)
- variance

variance = the model's confidence in each horizon

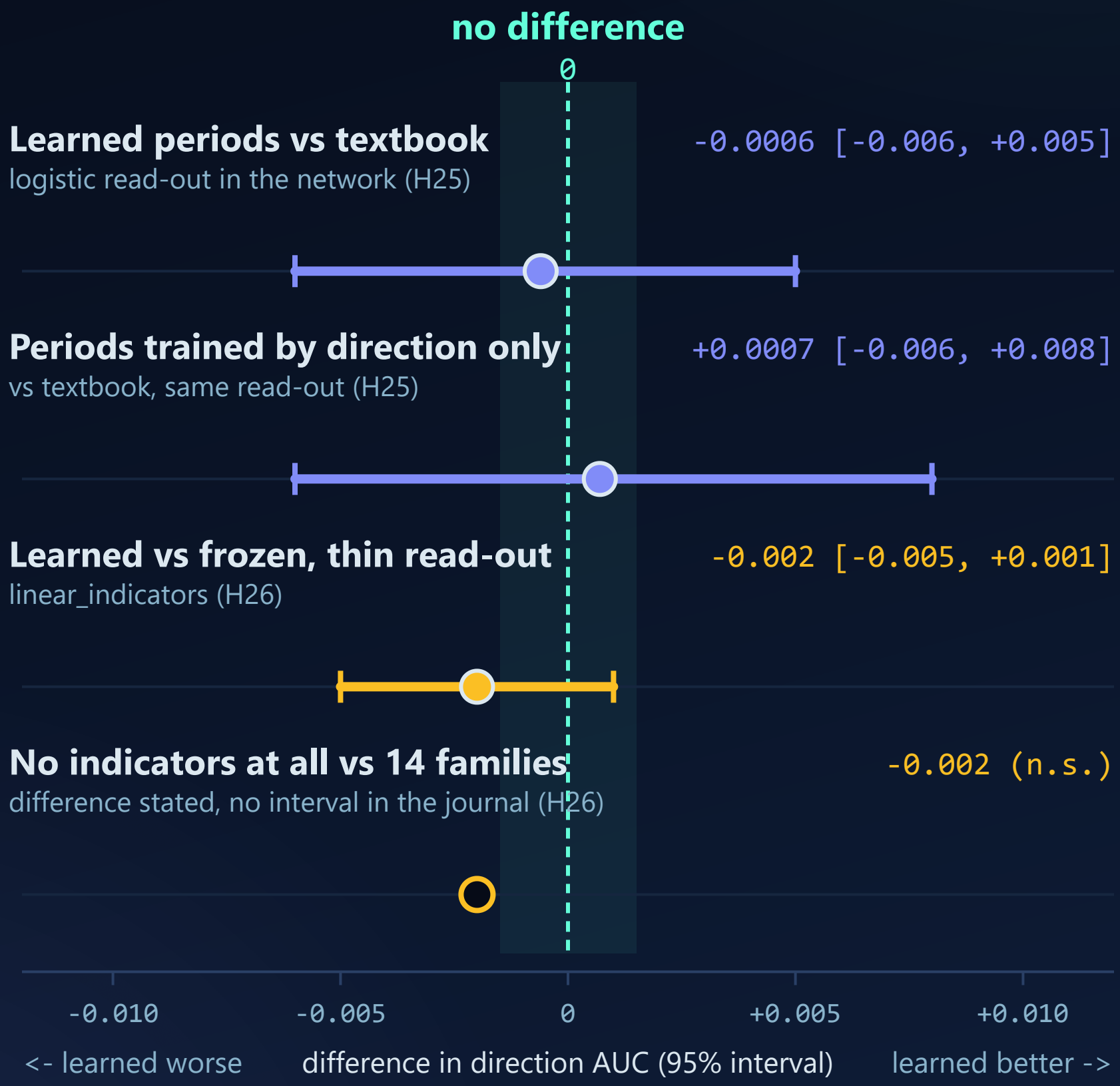
direction BCE
trains P(up)

Gaussian NLL + CRPS
train confidence

physics-inspired
T-perp, Casimir, HD, IFE, vacuum

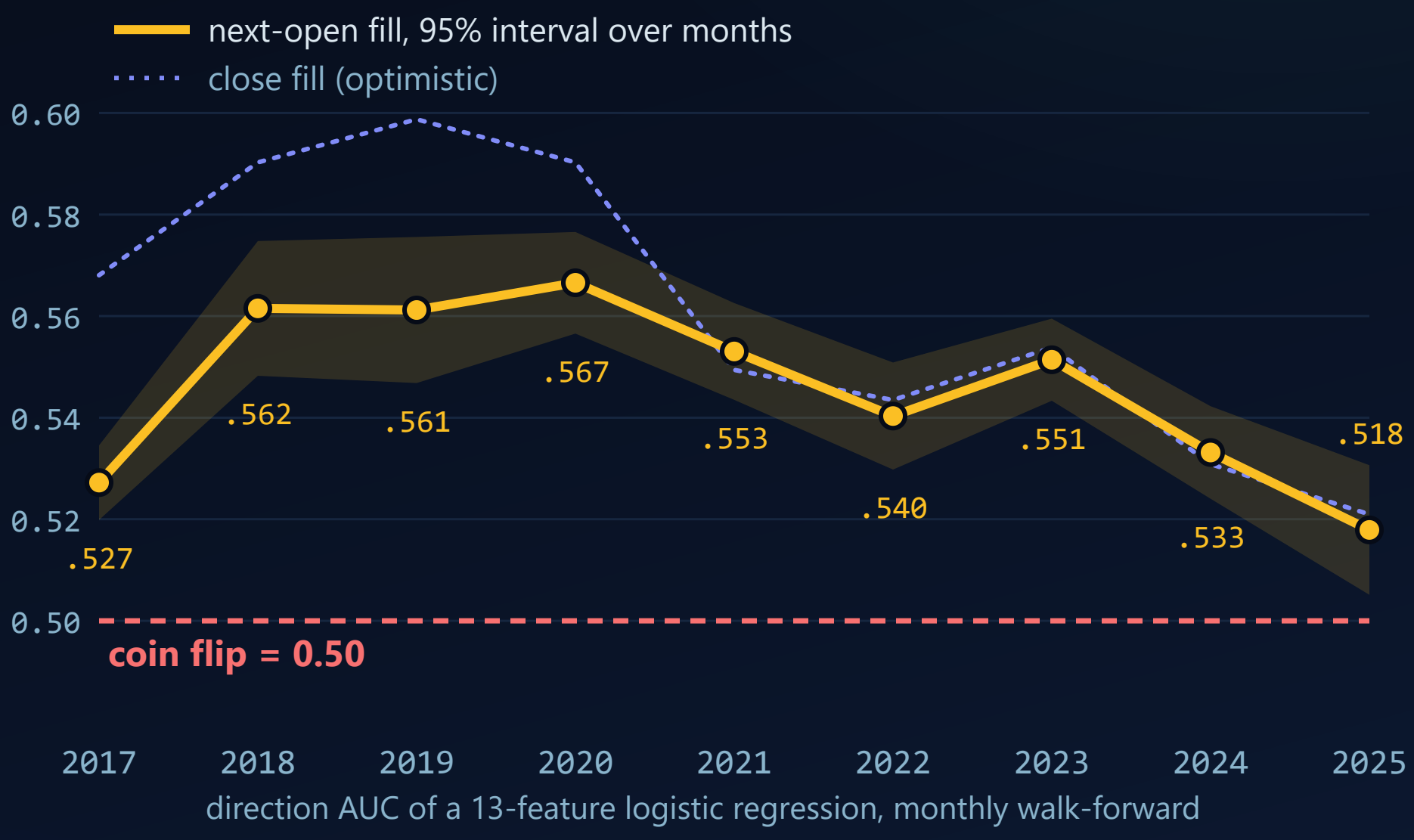
Learned periods = textbook periods

Did the network find better indicator settings than the textbook? Every interval shown crosses zero.



Even the best simple model drifted to a coin flip

The network scored lower still.



0.521 vs 0.570

network vs logistic regression on the same long blocks (H23)

0.535 vs 0.552

network vs logistic on lags, h1 AUC, 5 judgement folds (NT-104)

50.3% hit rate

forward test, 11 unseen months, 437 trades (H82)

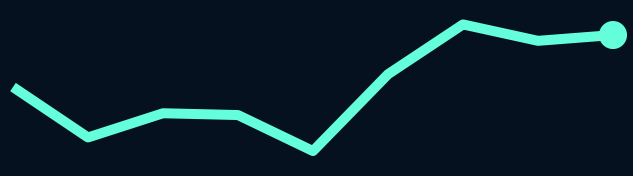
A price is a 1-D shadow of a multimodal world

So predicting it from past prices alone makes little sense.



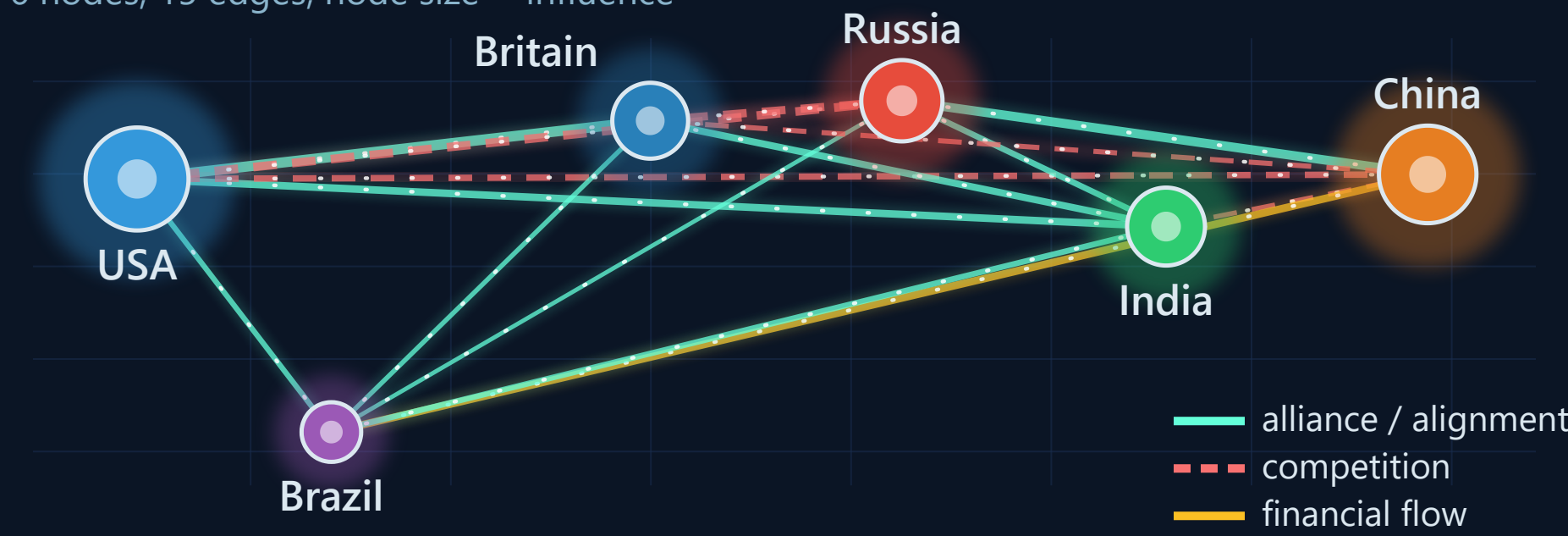
The **size** of the next move kept structure while its **sign** faded.

Volatility predictability (Spearman, HAR model): `0.286` in 2018, `0.399` in 2025 (peak 0.411 in 2023).



2017 - 2025

Five blocks of the world, one shared tensor



temporal-pattern
TFT forecasts of country ETFs

semantic-embedding
BERTopic over news

structural-centrality
geopolitical graph

dynamic-agent
agent simulation: resources, alliance eigenvalues

geospatial-kernel
10 raster layers at the 6 nodes

one 80-dimensional Universal Knowledge Tensor

stripes = block provenance labels, not to scale

SVD kernels → **sparse-autoencoder concepts** → **semantic canvas + governance flags**

Data: 32+ keyless public APIs (docs/ARCHITECTURE.md).
Political block: political-science datasets and a geopolitical graph.

I do not know if it will work.
I am optimistic.

NEXT INPUTS

- news — planned
- GIS layers — planned
- election results across democracies — planned
- shareholder votes across companies — planned

Simulation prototype

github.com/another-worlds/hyperspace

